

Cramer-Rao Lower Bound

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Following the book [Kay], we present some versions of Cramer Rao Lower Bound theorem.

1 Scalar version

Theorem 1. *Given a probability distribution function (PDF) $p(x; \theta)$ satisfy the "regularity" condition*

$$E \left[\frac{\partial \ln p(x, \theta)}{\partial \theta} \right] = 0, \forall \theta, \quad (1)$$

where the expectation is taken with respect to $p(x; \theta)$. Then, the variance of any unbiased estimator $\hat{\theta}$ must not lower than

$$\text{Var}(\hat{\theta}) \geq \frac{1}{-E \left[\frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right]}, \quad (2)$$

where the derivative at the true value of θ and the expectation is taken with respect to $p(x; \theta)$. Furthermore, the equality occurs for a specific unbiased estimator $\hat{\theta} = g(x)$ if and only if

$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = I(\theta)(g(x) - \theta), \forall \theta, \quad (3)$$

for some function g and I . $\hat{\theta} = g(x)$ is called MVU estimator and the minimum variance is $\frac{1}{I(\theta)}$.

Proof. □

Remark 1. *The expectation from (1) and (2) is determined as*

$$E \left[\frac{\partial \ln p(x, \theta)}{\partial \theta} \right] = \int \frac{\partial \ln p(x, \theta)}{\partial \theta} p(x; \theta) dx,$$

and

$$E \left[\frac{\partial^2 \ln p(x, \theta)}{\partial \theta^2} \right] = \int \frac{\partial^2 \ln p(x, \theta)}{\partial \theta^2} p(x; \theta) dx.$$

2 Vector version

Theorem 2. *It is assumed that the PDF $p(x; \theta)$ satisfies the "regularity" conditions*

$$E \left[\frac{\partial \ln p(x, \theta)}{\partial \theta} \right] = 0, \forall \theta, \quad (4)$$

where the expectation is taken with respect to $p(x; \theta)$. Then, the covariance matrix of any unbiased estimator $\hat{\theta}$ satisfies

$$C_{\hat{\theta}} - I^{-1}(\theta) \geq 0, \quad (5)$$

where ≥ 0 means that the matrix is positive semidefinite. The Fisher information matrix

$$[I(\theta)]_{ij} = -E \left[\frac{\partial^2 \ln p(x; \theta)}{\partial \theta_i \partial \theta_j} \right]$$

where the derivative are evaluated at the true value of θ and the expectation is taken with respect to $p(x; \theta)$. Furthermore, an unbiased estimator that makes $C_{\hat{\theta}} = I^{-1}(\theta)$ if and only if

$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = I(\theta)(g(x) - \theta), \quad (6)$$

for some p -dimensional function g and some $p \times p$ matrix I . $\hat{\theta} = g(x)$ is MVU estimator and its covariance matrix is $I^{-1}(\theta)$.

Proof. □

References

[Kay] *Fundamentals of Statistical Signal Processing, Volume 1: Estimation Theory*. Pearson Education.